

Juan Esteban Sereno Mesa

PERSONAL INFORMATION

Name: Juan Esteban	Current Position: Postdoctoral Research Fellow
Last name: Sereno Mesa	Workplace: University of Idaho - Brink Hall 304
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Marital Status: Single	Phone: +1(208)9979055
Date of Birth: January 7, 1993	Email: jserenom@uidaho.edu, jserenom@gmail.com.
Place of Birth: Puerto Nare	ORCID: 0000-0002-8498-1277
Nationality: Colombian	Google Scholar: Juan E. Sereno

AREAS OF INTEREST

Mathematical Modelling, Model Predictive Control, Adaptive Control, Nonlinear Mixed-Integer Optimization, Infectious Diseases, Diabetes, Digital Twins.

ACADEMIC APPOINTMENTS

Postdoctoral Fellow, University of Idaho (September/2024 - Present), Department of Mathematics and Statistical Science, Moscow, Idaho, USA. Research areas: Adaptive Digital Twins for personalized medicine. In the SMID Lab under supervision of Prof. Esteban Hernandez-Vargas.

Doctoral Scholarship, Universidad Nacional del Litoral (April/2018 - April/2024), Institute of Technological Development for the Chemical Industry, Santa Fe La Capital, Santa Fe, Argentina. Research areas: Design of MPC controllers for switched systems adapted to realistic epidemiological outbreak scenarios. In the Dynamical System and Control Group, under supervision of Prof. Alejandro Hernán González.

EDUCATION

Ph.D. in Mathematical Engineering, Universidad Nacional del Litoral, Facultad de Ingeniería Química (2024), Argentina. Thesis Topic: “Design of Predictive Controllers for Hybrid Systems: Application to Socio-Biomedical Processes”. Advisor: Alejandro Hernán González.

M.Sc. in Engineering - Industrial Automation, Universidad Nacional de Colombia, Facultad de Minas (2021), Colombia. Thesis Topic: “Automatic Personalization of Glycemic Control Strategies for Patients with Type 1 Diabetes Mellitus”. Advisor: P. S. Rivadeneira.

B.E. in Control Engineering, Universidad Nacional de Colombia, Facultad de Minas (2017), Colombia. Final Project: “Glycemic Control in Patients with Type 1 Diabetes Mellitus”. Advisor: P. S. Rivadeneira.

AWARDS

Doctoral Internal Scholarship awarded by the National Council of Scientific and Technical Research (CONICET), Buenos Aires, Argentina, 2018.

First Place in the Area of Appropriate Technologies at the National Competition of Best Final Degree Projects Otto de Greiff, 21st edition, Bogotá - Colombia, 2017.

Best Final Degree Project at Universidad Nacional de Colombia, Medellín campus, Facultad de Minas, 2017.

TEACHING EXPERIENCE

Teaching Assistant	2022-2024
Teaching Assistant in the Mathematics Department, Facultad de Ingeniería Química, Universidad Nacional del Litoral, teaching Mathematics A.	

Teaching Assistant	2014 - 2015
Teaching Assistant in the Department of Electrical and Automatic Energy, Facultad de Minas, Universidad Nacional de Colombia, teaching Analog Electronics II.	

RESEARCH
PROJECTS

PICT 2022, Nro. 2021-I-A-00544.

2022 - 2024

Mathematical modeling and control of tuberculosis in Argentina. Funded by ANPCYT. Principal Investigator: Alejandro H. González, ARS \$6.000.000,00. UNL-CONICET, Santa Fe, Argentina. My Role: Doctoral Student.

PIP 2021, KE3 Nro. 112-2020010-2555-CO.

2021 - 2024

Mathematical modeling and control of the COVID-19 pandemic in Argentina. Funded by CONICET. Principal Investigator: Alejandro H. González, ARS \$1,510,000.00. UNL-CONICET, Santa Fe, Argentina. My Role: Doctoral Student.

PICT 2019, Nro. 2019-01794.

2019 - 2024

Design of economic model predictive control (EMPC) strategies for tracking time-varying trajectories and paths in the presence of obstacles. Funded by ANPCYT. Principal Investigator: Alejandro H. González, ARS \$1,063,125.00. UNL-CONICET, Santa Fe, Argentina. My Role: Doctoral Student.

JOURNAL
ARTICLES

1. **Sereno, J. E.**, Neujahr, Havilah, Hernandez-Gonzalez, Miguel, & Hernandez-Vargas, E. A. (2025). Iterative calibration of medical digital twins via adaptive estimators. *Frontiers in Applied Mathematics and Statistics*, 11, 1699390.
2. **Sereno, J. E.**, D’Jorge, A., Hernandez-Vargas, E. A., & González, A. H. (2025). Time-Constrained MPC Approach for Switched Nonlinear Systems With Applications to Biomedical Problems. *IEEE Control Systems Letters*, 9, 949-954.
3. **Sereno, J. E.**, D’Jorge, A., Ferramosca, A., Hernandez-Vargas, E. A., & González, A. H. (2023). Switched NMPC for epidemiological and social-economic control objectives in SIR-type systems. *Annual Reviews in Control*, 56, 100901.
4. **Sereno, J.**, Ferramosca, A., González, A., & D’Jorge, A. (2023). Impacts of Quantifying Social Distancing Measures on MPC Performance for SIR-Type Systems. *Latin American Applied Research*, 53(4), 417-422.
5. **Sereno, J.**, Anderson, A., Ferramosca, A., Hernandez-Vargas, E. A., & González, A. H. (2022). Minimizing the epidemic final size while containing the infected peak prevalence in SIR systems. *Automatica*, 144, 110496.
6. **Sereno, Juan E.**, D’Jorge, A., Ferramosca, A., Hernandez-Vargas, E. A., & González, A. H. (2021). Model predictive control for optimal social distancing in a type SIR-switched model. *IFAC-PapersOnLine*, 54(15), 251-256.
7. Godoy, J. L., **Sereno, J. E.**, & Rivadeneira, P. S. (2021). Meal detection and carbohydrate estimation based on a feedback scheme with application to the artificial pancreas. *Biomedical Signal Processing and Control*, 68, 102715.
8. Rivadeneira, P. S., **Sereno, J. E.**, Caicedo, M. A. (2019). Nuevas estrategias de control glucémico en pacientes con diabetes mellitus tipo 1. *Revista Iberoamericana de Automática e Informática*, 16(2), 238-248.
9. **Sereno, J. E.**, Caicedo, M. A., Rivadeneira, P. S. (2018). Artificial pancreas: glycemic control strategies for avoiding hypoglycemia. *Dyna*, 85(207), 198-207.
10. Rivadeneira, P. S., **Sereno, J. E.**, Magdelaine, N., Moog, C. H. (2017). Blood glycemia reconstruction from discrete measurements using an impulsive observer. *IFAC-PapersOnLine*, 50(1), 14723-14728.

BOOK
CHAPTERS

1. Rivadeneira, P. S., Godoy, J. L., **Sereno, J. E.**, Abuin, P., Ferramosca, A., González, A. H. (2020). Impulsive MPC schemes for biomedical processes: Application to type 1 diabetes. In *Control Applications for Biomedical Engineering Systems* (pp. 55-87). Academic Press.

CONFERENCE
PAPERS

1. Rodriguez-Liñan, Angel, **Sereno, J. E.**, Quiroz-Compean, Griselda, & Hernandez-Vargas, E. A. (2025). Adaptive Observers for Calibrating Immune Digital Twins in Viral Infections. 64th IEEE Conference on Decision and Control, 2025. Rio de Janeiro - Brazil.
2. **Sereno, J. E.**, D’Jorge, A., Hernandez-Vargas, E. A., & González, A. H. (2025). Time-Constrained MPC Approach for Switched Nonlinear Systems With Applications to Biomedical Problems. To be presented in the 64th IEEE Conference on Decision and Control, 2025. Rio de Janeiro - Brazil.
3. Rivadeneira, P. S., Caicedo, M. A., **Sereno, J. E.** (2018). A new Approach in Zone Model Predictive Control for Type 1 Diabetes to be Tested in Colombia. In Diabetes Technology & Therapeutics (Vol. 20, pp. A86-A87), Vienna - Austria.
4. **Sereno, J. E.**, González, A. H., & Rivadeneira, P. S. (2017, October). A performance comparison between standard and impulsive zmpp on type 1 diabetic patients. In 2017 IEEE 3rd Colombian Conference on Automatic Control (CCAC) (pp. 1-6). IEEE.
5. **Sereno, J. E.**, Bolaños, F., Vallejo, M. (2016). Artificial vision system for differential multiples robots. In 2016 Technologies Applied to Electronics Teaching (TAE) (pp. 1-5). IEEE, Sevilla - España.
6. Mohammadridha, T., Rivadeneira, P. S., **Sereno, J. E.**, Cardelli, M., Moog, C. H. (2016). Description of the positively invariant sets of a type 1 diabetic patient model. In 2016 Latin American Conference on Automatic Control (CLCA). IFAC, Medellín - Colombia.
7. **Juan Sereno**, Eliana I. Arango-Zuluaga, and Freddy Bolaños. (2014). Propuesta pedagógica para el desarrollo de un curso en robótica básica. III International Congress of Engineering Mechatronics and Automation, Cartagena - Colombia.
8. **Sereno, J. E.**, Ferramosca, A., González, A. H., & D’Jorge, A. (2023). Impacts of quantifying social distancing measures on MPC performance for SIR-type. 28° Congreso Argentino de Control Automático (AADECA) 2023, Buenos Aires - Argentina.
9. Abuin, P., **Sereno, J. E.**, Ferramosca, A., & Gonzalez, A. H.. Closed-loop MPC-based artificial pancreas: Handling circadian variability of insulin sensitivity. In 2020 Congreso Argentino en Control Automático (AADECA). October 2020, Buenos Aires - Argentina.
10. **Sereno, J. E.**, Caicedo, M. A., Rivadeneira, P. S., & Camacho, O. E.. In Silico Test for MPC and SMC Controllers Under Parametric Variations in Type 1 Diabetic Patients. In 2018 Congreso Argentino en Control Automático (AADECA). November 2018, Buenos Aires - Argentina.
11. **Sereno, J. E.**, Rivadeneira, P. S.. Auto-tuning for model predictive controllers in patients with type 1 diabetes. In 2018 Congreso Argentino en Control Automático (AADECA). November 2018, Buenos Aires - Argentina.

ACADEMIC
REFERENCES

Associate Professor Esteban A. Hernandez-Vargas, Department of Mathematics and Statistical Science, University of Idaho, USA, e-mail: esteban@uidaho.edu.

Professor Alejandro Hernán González, Facultad de Ingeniería Química - Universidad Nacional del Litoral, Principal Researcher at Instituto de Desarrollo Tecnológico para la Industria Química - CONICET, e-mail: alejgon@gmail.com.

Professor Antonio Ferramosca, Department of Management, Information, and Production Engineering, University of Bergamo e-mail: antonio.ferramosca@gmail.com.